

Jayanti Samaddar

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Education

- **Cornell University** Ithaca, NY
Master of Engineering in Biomedical Engineering 2023
Relevant Courses- Cardiac electrophysiology, Innovation & design of biomedical technologies (FDA, EU MDR, Regulatory & Quality focus), Management Cases, Management Consultancy, Product Management
- **Heritage Institute of Technology** Kolkata, India
B.E. Applied Electronics and Instrumentation Engineering 2021
Relevant Courses- Biomedical Instrumentation, Control Systems, Sensors & Transducers, Computational Biology, Optoelectronics

Experience

- **Tops Education** Remote, Beijing
Consultant Nov 2024- Present
Work in partnership with educators and advisors to prepare students for their applications to graduate programs. Review and refine personal statements, statements of purpose, and additional essays for universities in the United States & Canada.
Provide guidance and support to candidates applying for Master's and PhD programs.
Assist students in enhancing their academic profiles and application materials to increase their chances of admission.
Serve as a subject matter expert for students interested in a career in biomedical engineering.
- **Dream Venture Labs** Boston, MA
Research Intern in Search and AI Nov 2023- July 2024
Perform in-depth research to develop new search algorithms and AI applications.
Analyze large datasets to improve search functionalities and user experience. Collaborate with the Search and AI team to experiment with new search models and AI techniques. Worked on NSF Grant proposal.
- **Nationwide Children's Hospital** Columbus, OH
Associate Researcher Jan- Feb 2024
Analyzed sleep EEG data using python and MATLAB, studied slow wave oscillations on Global, local and frontal areas of the brain during deep sleep and naps. Performed background research and Literature reviews of articles in the topic. Aided Clinical Research coordinators with their study. Solved technical issues related to clinical studies.
- **Cornell University** Ithaca, NY
Design Engineer Aug 2022- May 2023
Collaborated with surgeons to understand unmet needs, analyzed gaps in current research
To better understand product criteria, created models for better visualization, led Scrum meetings, Managed progress using Notion.

Skills

Technical: C programming, Python, SQL, Microsoft Office Suite, Google Suite, AutoCAD, Fusion360

Management: Agile Methodologies, Kanban, Lean Six Sigma Green Belt

Research

- Design Engineer on creating a prototype for a synthetic amniotic sac through Cornell and under the supervision of our sponsor Dr. Rachelle Abdelnour, a maternal-fetal fellow at the University of Virginia
 - Researched embryonic development and complications in pregnancy with a special focus on the pre-term premature rupture of membranes.
 - Performed background research and arranged meetings with OB-GYN to understand current needs.
 - 3D printed sac deployment mechanism models for visualization using Fusion360.
 - Developed a solution to deploy a silicone sac around the fetus and laid the foundation for further study and development.
- Worked on creating a temperature-sensing device that uses multiple sensors to provide a wide range, functional in remote areas.
 - Led a team of five engineers, arranged scrum meetings to discuss project goals and requirements.
 - Wrote the code for Arduino to integrate it with RTD and Thermocouple.
 - Worked on creating a monitoring system that sends an alarm when the temperature fluctuates in the required range.

Professional development

- GCP & Biomedical responsible conduct of research certifications by CITI Feb 2024
- **Internet of Things (IoT) by Internshala** Kolkata, India
Training 1 month
Built an IoT Monitoring System, controlling devices over the Internet using, Cloud, APIs, Alerts, and Machine Learning.
- **Introductory Human Physiology by Duke University** October 2021
Online Course on Coursera

Membership

- **Instrument Society of India, IISc Bangalore** June 2018 – June 2019
Arranged seminars, workshops, and other student chapter activities.

Volunteer Work

- FIFA U-17 WORLD CUP INDIA 2017 October 2017
Managed their social media promotion, posted updates on the event, got people to sign up for events and kept track of daily activities.

Other

Languages- English (Fluent), Bengali (Native), Hindi (Fluent), and Japanese (Intermediate)